

E5 — independent falsification (bounded; cannot prove universals)

Scripts (auditor-written): `e5_exact.py` (Fractions/sympy; 74 checks incl. 5 negative controls; 7 s) and `e5_search.py` (pulp/CBC exact ILP, scipy/HiGHS LP, networkx; 13 checks + 1 runtime record, incl. 4 negative controls; 243.5 s; seed 20260928). Results: `e5_exact_results.json`, `e5_search_results.json`; logs `10_LOGS/E5_*.log`. Kill-switches were predeclared in the check names (a single FAIL would suspend the corresponding claim).

Results: 0 failures; every negative control failed as required.

Exact arithmetic covered: (4.6a) at $4 \cdot 10^{12}$ (and failure at $3.6 \cdot 10^{12}$, crossing $\approx 3.6725 \cdot 10^{12}$); Lemma 4.2 coefficient comparisons; all coefficients of Lemma 5.3 / (5.15) / (5.15a); first-phase constants 1600/2920/219; (5.12a), (5.13) (value $3.930083 \geq 3.93$; 3.931 fails), (5.14) ($0.10159 \leq 0.11$; $0.02783 \leq 0.029$); Q_0 derivation; Lemma 5.1 window numerics; (5.1a) regime inequalities symbolically; completing-the-square identity; (6.3) for all $0 \leq k \leq n < 300$; (6.4) and $\max_k f(n,k) = M(n)$ for $6 \leq n < 400$; (6.12); (E.3) for all $n \geq 1$, $s < 20$; (E.5); (E.2e) both inequalities; $\lfloor (2n+1)^2/24 \rfloor = M(n)$ ($n < 2000$) with the rational envelope as negative control; D.1/D.2 arithmetic and Table 8 residue coverage; tightness of 16 and 48 for the retained account (6.7) by LP (47 fails); (6.7a) closed form and its zero set $\{(1,1), (2,1)\}$; (5.18) expansion and error bound; Prop 8.1 contradiction chain and (8.5); C.3 schedule values a, z, k_0, h ($= \lfloor 4/(\delta/8)^5 \rfloor$), L_T, M_T .

Bounded searches: complete-split $cp = c_3 = c_4 = kh - C(k,2)$ and $W^* = 2C(k,2)$ for $2 \leq k \leq 5$, $k \leq h \leq 7$ (negative control $K_5 \vee I_2$ outside the hypothesis fails); $c_4(K_n) \leq M(n)$ for $n \leq 10$; $c_3(K_{10}) = 19 > 18$; **all 531 chordal graphs with ≤ 7 vertices satisfy $c_4(G) \leq M(n)$** (informational: consistent with $b = 0$ at these orders — the paper does not claim $b = 0$); transport inequality (4.4), F_4 monotone direction and chordality preservation on 120 random chordal instances (dropping the averaging factor fails, as it must); identity (6.7b) on 60 exact optimal partitions with random clique roots; E.3 witness has rooted defect $\leq s$, not $\leq s-1$, and $cp = Q_s(n)$ ($s \leq 2$); `RootedDefectAt 0  $\Leftrightarrow$  chordal` for all graphs on ≤ 6 vertices.

Certo certificate re-examination (mandate E5). Not performed: it lives in the internal-audit package, whose contents are read only after the independent first pass (E6 incomplete) and after the E7 blocker stopped new work. → deferred.

Limitations. Small orders only; the asserted theorems are asymptotic with astronomically large thresholds, so no bounded test can refute Theorems B, C, 6.1 or Cor 6.1a directly. The tests exercise the identities, constants and mechanisms on which they rest.

E5 = INCONCLUSIVE — every executed test passes and every negative control fails as required (no counterexample), but the mandated Certo-certificate re-examination was not performed.

Rendered from the Markdown semantic source with PyMuPDF; not compiled from the TeX file (no TeX engine installed).